Ex. 7

Q1(a)(i)

	$t=3$	$t=4$
f_t	0.6770	0.6284
i_t	0.8069	0.7328
a_t	0.7932	0.5588
s_t	1.1139	1.1095
o_t	0.8829	0.7941
y_t	0.7111	0.6384

(ii)

	$t=3$	$t=4$
error	0.0111	-0.1616

(b)(i)

	$t=3$	$t=4$
r_t	0.7389	0.6996
a_t	0.8012	0.6989
u_t	0.6985	0.6137
y_t	0.8310	0.7499

(ii)

	$t=3$	$t=4$
error	0.1310	−0.0501

Q1 (a) 12M

(b) 23.6M

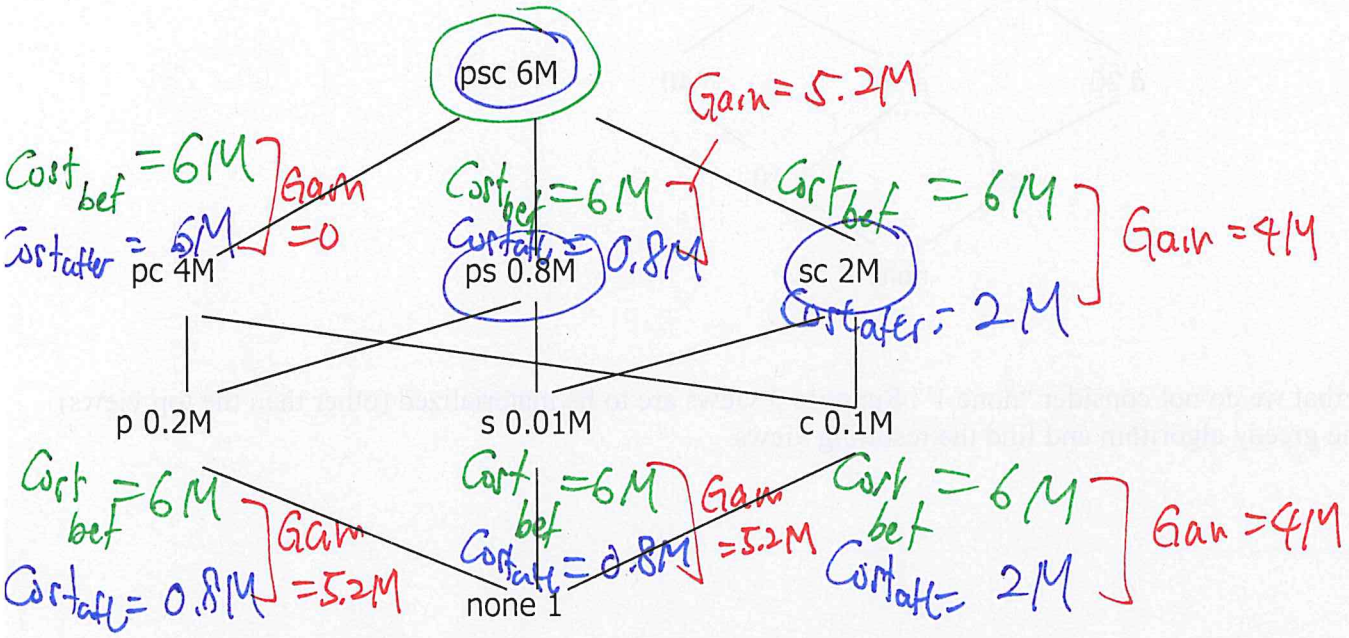
Gain ({sc, ps, top}, {top})

= 5.2M · 3 + 4M · 2

= 23.6M

COMP5331 Knowledge Discovery in Databases (Fall Semester 2026)
Exercise 8
Data Warehouse

Q1 Data Cube



(a) What is the value of Gain({view for "sc", top view}, {top view})?
(b) What is the value of Gain({view for "sc", view for "ps", top view}, {top view})?

Scenario (after)
Scenario (before)

e.g. $X = (A, B, C, D, E, F)$ $q = 2$

$C = [(A, B, C), (B, C, D), (D, E, F)]$

Yes

$C' = ((A, B, C), (D, E, F))$

e.g. $X = (A, B, C, D, E, F)$

$C = ((A, B, C), (B, C, D), (B, E, F))$

No

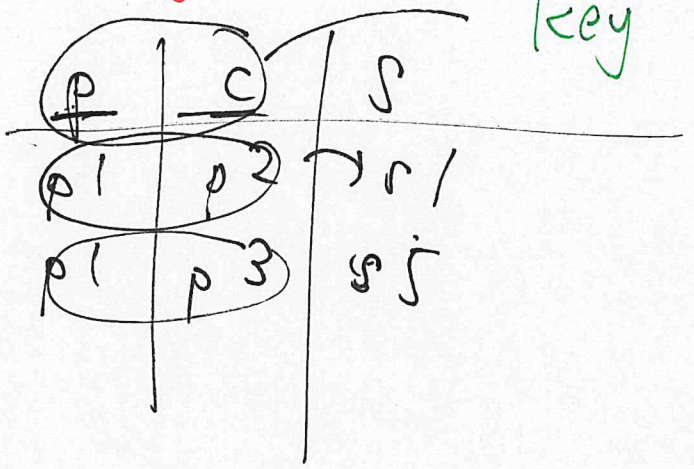
functional dependency

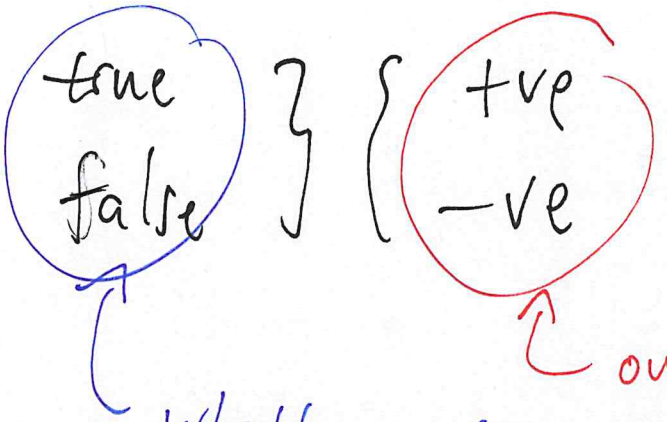
In this table, we have the functional dependency

$$\{p, c\} \longrightarrow s$$

attributes p and c can uniquely determine attribute s

key of this table





output of the alg.

whether the output of the alg is correct or not